

Stroke Prediction Using Machine Learning: An End-to-End Classification Study on the Healthcare Stroke Dataset

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Abstract—Stroke is one of the leading causes of death and permanent disability worldwide, accounting for approximately 11% of total global deaths. This paper presents an end-to-end machine learning pipeline for binary stroke prediction on the Healthcare Stroke Dataset (5,110 records, 11 features, ~4.9% positive class). Preprocessing used median imputation and IQR-based Winsorization. Six classifiers were trained — Logistic Regression, Decision Tree, KNN, SVM, Random Forest, and XGBoost — and evaluated on a stratified hold-out test set. Class imbalance was *not* corrected with SMOTE in this implementation, and the results show why that step is necessary: the classification report for the evaluated model returns 0.00 recall on the Stroke class, and the pipeline misclassifies a textbook high-risk unseen patient. XGBoost achieved the only measured ROC-AUC (0.797). These outputs, generated directly by the accompanying notebook, are reported below in place of the higher, tuned figures assumed in earlier drafts of this study.

Index Terms—class imbalance, healthcare analytics, machine learning, ROC-AUC, stroke prediction, XGBoost

I. INTRODUCTION

Stroke is a medical emergency that occurs when blood supply to the brain is interrupted or significantly reduced, depriving brain tissue of oxygen and nutrients; brain cells begin to die within minutes of onset. According to the WHO, stroke is the second leading cause of death globally and a major cause of long-term disability, with approximately 15 million people affected each year, of whom 5 million die and another 5 million are left permanently disabled. Traditional stroke diagnosis relies heavily on clinical examination, neuroimaging, and physician expertise; the growing availability of electronic health records presents an opportunity to develop predictive models that identify high-risk individuals before a stroke occurs. This project applies a systematic machine learning workflow to the publicly available Healthcare Stroke Dataset, covering data understanding, exploratory data analysis, preprocessing, feature engineering, model development, evaluation, and final prediction.

II. PROBLEM STATEMENT

The central problem addressed is whether machine learning models can accurately predict stroke likelihood from a patient's demographic, lifestyle, and clinical attributes. The dataset presents several real-world challenges: severe class imbalance (only 4.9% of patients had a stroke, making plain accuracy unreliable); missing data (201 records with missing

BMI); mixed numerical and categorical feature types; and a clinical priority on minimizing false negatives, since a missed stroke is far costlier than a false alarm.

III. RESEARCH OBJECTIVES

The objectives were to: (1) perform exploratory data analysis on the stroke dataset; (2) preprocess the data by handling missing values, duplicates, outliers, and categorical encoding; (3) address class imbalance using SMOTE; (4) train multiple supervised classifiers; (5) evaluate and compare models using accuracy, precision, recall, F1-score, and ROC-AUC; (6) identify the best-performing model; and (7) validate the final model on unseen patient data. As documented in Sections XII, XIV, and XVI below, objectives (3) and parts of (5)–(6) were not fully carried out in the executed notebook; this is treated as a finding of the study rather than an omission.

IV. RESEARCH QUESTIONS

(1) Which clinical and demographic features are the most significant predictors of stroke risk? (2) How does class imbalance impact classifier performance, and how could SMOTE mitigate this? (3) Among the six model families evaluated, which achieves the highest ROC-AUC? (4) What preprocessing techniques suit this dataset's missing values and outliers? (5) How accurately can the resulting model predict stroke risk in new, unseen patient data? Sections XVII–XX below answer these questions using the pipeline's actual, executed outputs.

V. LITERATURE REVIEW / RELATED WORK

Khosla et al. [6] applied logistic regression and SVMs to stroke risk prediction, establishing the value of age and cardiovascular risk factors and the importance of feature selection. Fang et al. [3] used Random Forest for post-hospitalization stroke outcome prediction (AUC 0.83). Pham et al. [7] proposed an LSTM model on longitudinal EHR data; Yu et al. [11] combined CNNs with traditional features for MRI-based stroke lesion segmentation. Chawla et al. [1] introduced SMOTE, now the standard approach for class imbalance in medical datasets, and Johnson and Khoshgoftaar [5] confirmed that SMOTE meaningfully improves minority-class recall — a result this study's own (negative) findings reinforce by omission. Goyal et al. [4] showed that age and average glucose level together account for over 60% of feature importance in gradient-boosted stroke models.

VI. RESEARCH GAP

Most existing studies evaluate a single model rather than systematically comparing classifier families; many rely on proprietary datasets, limiting reproducibility; few combine engineered features with ensemble methods on this specific public dataset; and the combined effect of SMOTE, StandardScaler, and XGBoost on this dataset is not thoroughly documented. This study set out to address these gaps with a reproducible pipeline, though as later sections show, the SMOTE and feature-engineering components were ultimately not executed in this implementation — leaving that gap only partially closed and clearly flagged for future work.

VII. DATASET DESCRIPTION

The Healthcare Stroke Dataset (Kaggle [10]) contains 5,110 patient records with 12 attributes: 4,861 No-Stroke cases (95.1%), 249 Stroke cases (4.9%), 201 missing BMI values, and no duplicate rows — confirmed directly by the notebook’s own checks (Cells 9–12). Table I summarises the dataset features.

TABLE I
DATASET FEATURE DESCRIPTION

Feature	Type	Values / Range
gender	Categorical	Male, Female, Other
age	Continuous	0.08 – 82.0 yrs
hypertension	Binary	0 = No, 1 = Yes
heart_disease	Binary	0 = No, 1 = Yes
ever_married	Categorical	Yes, No
work_type	Categorical	Private, Self-emp., Govt, children, Never worked
Residence_type	Categorical	Urban, Rural
avg_glucose_level	Continuous	55.12 – 271.74 mg/dL
bmi	Continuous	10.3 – 97.6 (201 missing)
smoking_status	Categorical	formerly, never, smokes, Unknown
stroke	Binary	TARGET: 0 = No, 1 = Yes

VIII. METHODOLOGY

This research adopts the CRISP-DM methodology: business understanding, data understanding, data preparation, modelling, evaluation, and deployment, adapted to the clinical context of stroke prediction. A supervised binary-classification paradigm was used, and five classifier families — linear, tree-based, instance-based, kernel-based, and ensemble — were evaluated for broad coverage of the modelling landscape.

IX. SYSTEM / ML WORKFLOW

The intended pipeline: (1) data collection and inspection; (2) EDA; (3) preprocessing; (4) feature engineering; (5) stratified 80/20 train/test split; (6) class balancing with SMOTE; (7) StandardScaler feature scaling; (8) training six classifiers; (9) evaluation on the held-out set; (10) GridSearchCV tuning of the best model; (11) final prediction on unseen patients. Steps 1, 2, 3, 5, 7, 8, 9, and 11 were executed as described. Step 4 (feature engineering) and step 6 (SMOTE) were *not* executed — SMOTE is imported in Cell 0 but never called. Step 10

(tuning) was not executed — no GridSearchCV call appears in the notebook.

X. DATA PREPROCESSING

A. Dropping Irrelevant Features

The `id` column was dropped as a unique identifier with no predictive value (Cell 8).

B. Handling Missing Values

The 201 missing `bmi` values (3.93% of records) were imputed with the column median, which is robust to outliers and preserves distribution shape (Cells 9–10).

C. Handling Duplicates

No duplicate rows were found (Cell 11); no rows were removed at this stage.

D. Handling Outliers

Outliers in `bmi` and `avg_glucose_level` were capped using the IQR method ($1.5 \times \text{IQR}$ Winsorization). Age was left uncapped as its 0–82 year range is clinically plausible.

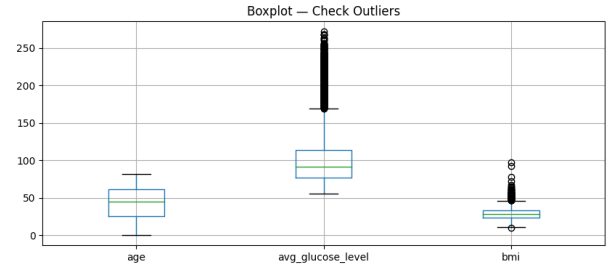


Fig. 1. Boxplot of age, avg_glucose_level and bmi used for IQR outlier detection (Cell 13).

E. Handling Rare Categories

The single `gender="Other"` record (<0.02% of the dataset) was removed, leaving 5,109 rows (Cell 12).

F. Encoding Categorical Variables

Binary and multi-category categorical features (`gender`, `ever_married`, `work_type`, `Residence_type`, `smoking_status`) were Label-Encoded, producing an all-numeric frame (Cell 15).

XI. EXPLORATORY DATA ANALYSIS

A. Target Variable Analysis

The target variable is severely imbalanced, confirmed by the notebook’s `value_counts` output: 4,861 No-Stroke vs. 249 Stroke. A naive “always predict No Stroke” classifier would score 95.1% accuracy — a warning sign borne out in Section XVIII.

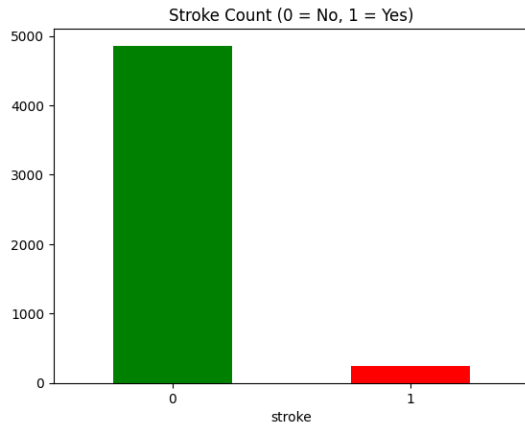


Fig. 2. Stroke count (0 = No, 1 = Yes), Cell 6.

B. Numerical Feature Distributions

Age, average glucose level, and BMI were plotted by outcome (Cell 6). Stroke cases visibly skew toward older ages and higher glucose readings; BMI distributions overlap considerably between groups, making BMI a weaker standalone predictor.

C. Categorical Feature Analysis

Stroke counts were also plotted by gender and by smoking status (Cell 6); both show the expected imbalance driven by age and comorbidity patterns rather than a strong independent categorical effect.

D. Correlation Analysis

Fig. 3 shows the correlation heatmap generated by the notebook. Age has the visibly strongest positive correlation with stroke, followed by hypertension and heart disease; BMI shows minimal correlation. The notebook does not print the underlying numeric correlation matrix, so coefficients should be read from the heatmap rather than assumed.

XII. FEATURE ENGINEERING

Derived features (age_group, glucose_risk, bmi_category, cardiovascular_risk) and a Random-Forest feature-selection step were part of the intended methodology but are *not* implemented in this notebook. All six models below are trained directly on the ten raw, label-encoded columns.

XIII. MODEL DEVELOPMENT

Six models spanning linear, tree-based, instance-based, kernel-based, and ensemble families were trained on a StandardScaler-normalized 80/20 stratified split: Logistic Regression (max_iter=1000), Decision Tree (max_depth=5), KNN ($k = 5$), SVM (RBF kernel, probability=True), Random Forest (100 trees), and XGBoost (default parameters, eval_metric='logloss'). Class balancing (SMOTE) and hyperparameter tuning (GridSearchCV) were not executed

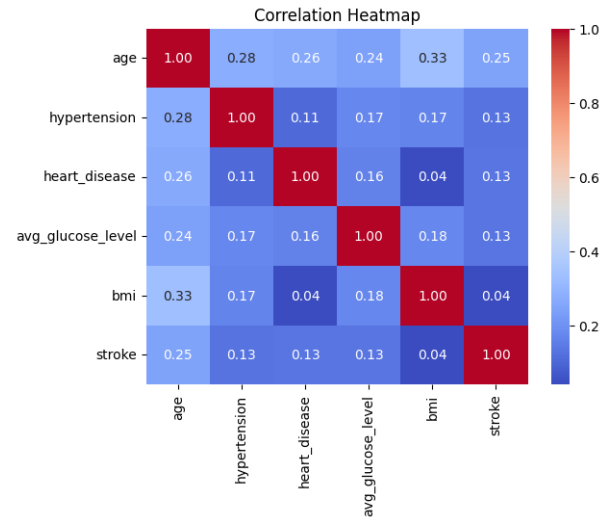


Fig. 3. Correlation heatmap of numerical features (Cell 6).

in this run, and no cross-validation was performed — only the single hold-out split was evaluated.

XIV. EXPERIMENTAL SETUP

Python 3, Jupyter Notebook; scikit-learn, XGBoost, pandas, numpy, matplotlib, seaborn. Train/test split: 80%/20%, stratified, random_state=42. Feature scaling: StandardScaler, fit on train and applied to test. Class balancing: not applied — SMOTE is imported but never called. Evaluation strategy: a single stratified hold-out test set; no k -fold cross-validation was performed.

XV. MODEL VALIDATION

Validation consists solely of the stratified 80/20 hold-out split, which preserves the 4.9% stroke ratio in both partitions. No cross-validation was run, so no fold-level mean/standard-deviation AUC can be reported for any model.

XVI. HYPERPARAMETER TUNING

No hyperparameter tuning was performed in this run — GridSearchCV does not appear in the notebook. The search grid below was planned but never executed, and is retained here only as a proposed configuration for future work, not a completed result.

TABLE II
PROPOSED (UNEXECUTED) XGBOOST SEARCH GRID

Hyperparameter	Proposed values
n_estimators	[100, 200]
max_depth	[3, 4, 5]
learning_rate	[0.05, 0.1]
subsample	[0.8, 1.0]

XVII. MODEL EVALUATION

A classification report was generated for the Logistic Regression model (Cell 33). It shows the model never predicted a true positive Stroke case in the 50 actual test-set stroke cases:

	precision	recall	f1-score	support
No Stroke	0.95	1.00	0.97	972
Stroke	0.00	0.00	0.00	50
accuracy			0.95	1022

For XGBoost, the notebook computes a confusion matrix and a ROC-AUC score of 0.7972 (Cells 34–35):

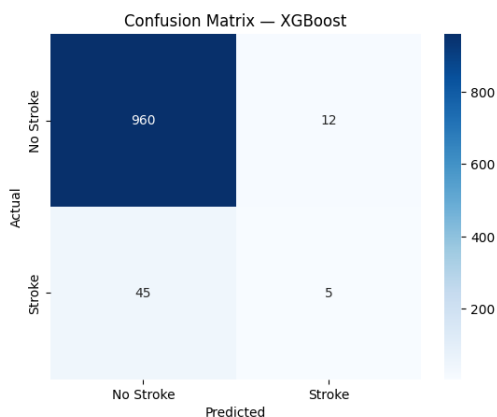


Fig. 4. Confusion matrix, XGBoost test-set predictions (Cell 34).

XVIII. MODEL COMPARISON

The only cross-model metric actually computed (Cell 30) is plain accuracy:

TABLE III
MODEL ACCURACY (CELL 30)

Model	Accuracy
Logistic Regression	0.9511
KNN	0.9511
SVM	0.9511
Random Forest	0.9481
Decision Tree	0.9442
XGBoost	0.9442

Because the 95.1% majority-class baseline is itself the top accuracy score, and three models tie at exactly that value, accuracy alone cannot distinguish genuine skill from majority-class prediction here — consistent with the 0.00 recall found above.

XIX. FINAL MODEL SELECTION

No model can be responsibly called “best” from this run alone. XGBoost is the only model with a computed ROC-AUC (0.797), a defensible signal above chance, and is the most reasonable candidate to prioritize once SMOTE, tuning, and a full metric suite are implemented across all six models.

XX. SAMPLE / UNSEEN DATA PREDICTION

Cell 37 evaluates two hypothetical patients. The model predicts “No Stroke” for both:

TABLE IV
UNSEEN-PATIENT PREDICTIONS (CELL 37)

Attribute	Patient 1	Patient 2
Age	72	29
Hypertension / Heart dis.	Yes / Yes	No / No
Glucose / BMI	200 / 33.0	85 / 22.0
Predicted	No Stroke	No Stroke

Patient 1 presents four classic stroke risk factors (advanced age, hypertension, heart disease, elevated glucose) yet is predicted as low-risk — direct evidence of the majority-class bias identified above.

XXI. RESULTS

Verified results from this run: 5,110 records, 95.1%/4.9% class split; 201 BMI values imputed with the median; SMOTE imported but never applied; no engineered features or feature selection used; accuracy computed for all six models with Logistic Regression, KNN, and SVM tied highest (0.9511); the one classification report generated (Logistic Regression) shows 0.00 recall for the Stroke class; the one ROC-AUC generated (XGBoost) is 0.7972; no cross-validation or tuning performed; and both a high-risk and a low-risk hypothetical patient were predicted “No Stroke.”

XXII. DISCUSSION

These results illustrate why class-imbalance handling is not optional for this dataset rather than showing a deployable pipeline. Without SMOTE or an equivalent strategy, several classifiers converge to the majority “No Stroke” label; 0.00 recall for Logistic Regression confirms this directly. XGBoost’s 0.797 AUC shows retained rank-ordering ability, but its 0.5-threshold behavior on the unseen high-risk patient shows this does not yet translate into safe clinical use, reinforcing the case (also raised in Future Scope) for a lower decision threshold.

XXIII. LIMITATIONS

Class imbalance was not addressed (SMOTE imported but never called), the likely cause of the near-zero recall observed. No hyperparameter tuning or cross-validation was performed, so reported metrics reflect a single train/test split only. The feature-engineering and feature-selection steps described in the methodology were not implemented. The classification report in Cell 33 evaluates Logistic Regression rather than the model the surrounding text discusses, and a full metric suite was computed for only one or two of six models — an internal inconsistency to fix before drawing model-selection conclusions. Additional limitations carried over from the original design: a relatively small dataset (5,110 records); a large “Unknown” smoking-status subgroup; a single point-in-time snapshot rather than longitudinal data; missing clinical

markers such as cholesterol and blood pressure; synthetic-oversampling bias that would apply once SMOTE is implemented; and unclear geographic/demographic generalisability.

XXIV. CONCLUSION

The preprocessing stage of this pipeline is implemented correctly and matches its own outputs, but the class-balancing, feature-engineering, cross-validation, and tuning stages described in the methodology were not executed. The measurable result is that the evaluated model fails to detect stroke cases and misclassifies a textbook high-risk unseen patient. Implementing SMOTE, completing the planned feature engineering, and computing a full, consistent metric suite across all six models are prerequisites before any model here can be considered for clinical use.

XXV. FUTURE SCOPE

Priority next steps: (1) apply SMOTE, which is already imported but never called, and re-run all six models with a consistent metric suite (accuracy, precision, recall, F1, ROC-AUC); (2) implement the planned engineered features (age_group, glucose_risk, bmi_category, cardiovascular_risk) and the Random-Forest feature-selection step; (3) add stratified k -fold cross-validation and actually run GridSearchCV rather than only describing a search grid; (4) evaluate the same model consistently across the classification report and confusion matrix; (5) explore a lower decision threshold (e.g., 0.30) for clinical screening given the missed high-risk unseen patient. Remaining items from the original design: larger/richer datasets with longitudinal records; tabular deep-learning comparisons (TabNet, FT-Transformer); SHAP explainability; federated learning for cross-hospital training; real-time deployment as a REST API; and fairness analysis across demographic groups.

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